

SALIM SADMAN

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Mechanical engineer who builds robots from the actuator up: mechanism design, fabrication, PCBs, firmware and motor control. Designed a custom cycloidal quasi-direct-drive (QDD) actuator and built the 12-DoF quadruped platform driven by twelve of them for a B.Sc. thesis on muddy-terrain locomotion, and co-led a university rover team to the finals of the Anatolian Rover Challenge 2026.

Research interests: Robot design and fabrication · Actuator design and motor control · Legged locomotion on deformable terrain · Field, space and marine robotics · Mechatronic system integration

EDUCATION

B.Sc. in Mechanical Engineering

Mar 2022 – Sep 2026

Rajshahi University of Engineering & Technology (RUET), Rajshahi, Bangladesh

CGPA: 3.56 / 4.00

Thesis: *Design and Development of a Quadruped Robot for Optimizing Muddy Terrain Locomotion*

Supervisor: Prof. Md. Emdadul Haque, Department of Mechanical Engineering, RUET

HSC (Science), Dhaka College, 2020, and **SSC (Science)**, B.L. Govt. High School, Sirajganj, 2017 — GPA 5.00 / 5.00 in both

RESEARCH & ENGINEERING EXPERIENCE

Quadruped Robot with Custom QDD Actuators for Muddy-Terrain Locomotion

Ongoing

B.Sc. thesis research, RUET: actuator design, mechanical design, fabrication and bench testing

- Designed a compact **quasi-direct-drive joint actuator**: BLDC motor, cycloidal reducer, encoder feedback and field-oriented control (FOC), for the torque density and backdrivability that legged locomotion needs
- Fabricated and assembled all twelve actuators and the robot's full **12-DoF structure** in-house
- Built a **motor-characterization test bench** (ODrive v3.6 + ESP32) whose Python pipeline time-aligns controller telemetry (velocity, current, power, faults) with output-side measurements to map efficiency
- Designed the platform to carry **IMU and depth-camera sensing** for locomotion experiments on mud and other deformable ground

MEER 2.0 Mars Rover, Team Ogrodooot

Oct 2024 – Sep 2026

Co-Lead & Mechanical Sub-Team Lead (Aug 2025 – Sep 2026); Mechanical Design & Fabrication (Oct 2024 – Aug 2025)

- Designed and fabricated the rover's mechanical system: **4-wheel rocker-bogie chassis** with differential suspension, custom wheels, and a **5-DoF robotic arm** with linear actuators and a motor-driven gripper
- Co-led the team to the **finals of the Anatolian Rover Challenge (ARC) 2026**, an international university rover competition, while heading its mechanical sub-team
- Ran **20+ mentoring sessions** (2025–26) for junior members on mechanical-team practice and CAD design in Fusion 360; closely involved in recruiting new members

Underwater Glider for Aquatic Environmental Monitoring

- Developed an underwater glider with **buoyancy-based depth control**, integrating dissolved-oxygen, ORP, turbidity and TDS sensors for water-quality monitoring

IoT Engineer (Hardware Engineering)

Jan 2024 – Jan 2025

SystemSage Solutions Sdn. Bhd.

- Designed and prototyped **ESP32-based PCBs** for data acquisition and automation
- Built **solar-powered monitoring and control nodes** for remote deployment: ambient (BME280) and soil-nutrient (RS485 NPK) sensing, relay actuation, and LoRa, Wi-Fi, GPRS and 4G LTE telemetry
- Co-authored a conference paper on the resulting smart-farming system, **published at GEn-CITY 2024 [P1]**

PUBLICATIONS

[P1] Mustahoshin Hossain Ahamed Afif, Javid Iqbal, **Salim Sadman Borshon**, Rashedul Islam Junayed, and Sabbir Ahamed, "An IoT-based smart farming solution for sustainable agriculture: integrating AI, cloud computing, and 4G communication for enhanced productivity," in *Proc. Int. Conf. on Green Energy, Computing and Intelligent Technology (GEn-CITY 2024)*, Malaysia (virtual), Dec. 2024. *IET Conference Proceedings*, vol. 2024, no. 30, pp. 83–88. doi: [10.1049/icp.2025.0238](https://doi.org/10.1049/icp.2025.0238) (also in IEEE Xplore)

TEACHING, MENTORING & OUTREACH

Robotics Instructor

Jan 2025 – Present

Artificial Intelligence Bangladesh (AIBD); full-time since Aug 2026

- Taught **80+ classes with 500+ student engagements** in robotics, electronics and programming, and helped organize **five regional and national robotics events**, initially alongside full-time undergraduate study
- Flagship programs: a 15-day robotics and 3D-printing workshop at Bangladesh Computer Council, Rajshahi; a 4-day robotic-hand design and control workshop at American Corner Rajshahi; a robotic-hand session for IEEE RAS, University of Rajshahi
- Wrote the organization's **course modules** — electronics, robotics, line-following robots, C programming, Python (including object-oriented programming) and, most recently, CAD and 3D printing — plus the illustrated assembly manual for its line-follower kit (custom PCB, Arduino, IR sensors, PID control)

Workshop Instructor and Organizer

2024 – 2025

Robotic Society of RUET

- Organized and taught three workshops (six classes, **200+ student engagements**) on robotics, electronics, microcontrollers and PCB design

LEADERSHIP & SERVICE

Robotic Society of RUET

Nov 2023 – Aug 2026

General Secretary (2025–26) · Assistant Technical Secretary (2024–25) · Assistant Technical Coordinator (2023–24)

- Secured sponsorship for and organized “Ro Te Robot”, a national robotics event; ran three annual recruitment drives

IEEE RUET Student Branch

Feb 2025 – Sep 2026

Sponsorship Coordinator (2025–26) · Content Writer (2025)

- Raised funding that launched several branch programs; ran a 2025 STEM outreach event at B.L. Govt. High School

HONORS & AWARDS

Finalist, Anatolian Rover Challenge (ARC), international university rover competition (Team Ogradoot) 2026

Semi-finalist, International Rover Design Challenge (IRDC), Space Robotics Society (Team Ogradoot) 2025

1st Place, Poster Presentation, RUET Innovista 2025

2nd Place, Project Showcase, Astro Science Fair, BSMR Novo Theatre, Rajshahi 2024

1st Place, Line-Following Robot Competition, Tech-Mayhem, Robotic Society of RUET 2023

Pre-university (2012–2017): Champion, Regional Math Olympiad; 1st Place, district Science Olympiad (National Science & Technology Fest); two 1st places, a 2nd and a 3rd in district, division and inter-college project competitions

TECHNICAL SKILLS

Mechanical design: Fusion 360 (including Python API scripting); mechanism design (cycloidal reducers, rocker-bogie suspension, lead screws, grippers); FDM 3D printing in ABS and TPU (Ender-3 V2/V3, Bambu Lab P1S)

Actuation & control: BLDC motors, field-oriented control (FOC), ODrive, encoder feedback, PID control, motor characterization and test-bench design

Embedded & electronics: ESP32, STM32, Arduino; PCB design (EasyEDA); IMU, depth-camera, RS485 and water-quality sensor integration; LoRa, Wi-Fi, GPRS/4G LTE; solar power systems

Programming: C/C++, Python (data acquisition and telemetry pipelines), MATLAB

TRAINING & INDUSTRIAL EXPOSURE

Industrial attachment — Walton Hi-Tech Industries PLC, Department of Engineering Services Management, Chandra, Gazipur (2–25 March 2025, 20 working days): studied high-volume manufacturing end to end, from refrigeration and HVAC assembly, PCB fabrication, battery packs, consumer electronics, fasteners and plastics recycling to the R&D section, questioning line engineers on process and design decisions throughout

Basic Programming with Python, 60-hour course (Dec 2024 – Feb 2025): IICT, RUET, under the EDGE project of Bangladesh Computer Council, ICT Division; fundamentals through introductory machine learning

Study tour (2025): 670 MW combined-cycle dual-fuel power plant, organized by IEEE RUET Student Branch